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The Fig. 1 phylogenetic tree was built using the GenBank query "Zika virus envelope protein (E) gene", selected because it exactly matches the top NCBI BLAST result. Fig. 2 presents a larger tree from the broader query "Zika virus envelope protein". I focus on Fig. 1 as it ensures full genomic sequences including the E gene and its context, whereas Fig. 2 may contain only a portion of the E gene that encodes for the envelope protein. Both trees are included for a broader perspective.

Discussion

Fig. 1 shows strain clustering by country of origin, with Fig. 1a and Fig. 1b representing the same tree in different layouts. Branch lengths in Fig. 1a highlights genetic similarities, with most strains from the same region clustering together, suggesting localized outbreaks and regional evolution. Additionally, 23 of the 30 strains originate from Latin America (Brazil, Cuba, Mexico, and Ecuador), while 6 are from Asia (India and Thailand), and one has an unknown origin (AF372422.1). Given the patient's recent travel to Brazil, the predominance of Latin American strains makes sense.

In the NCBI BLAST results, the top match was strain AF372422.1, but its origin is not available. I attempted to access its source article, but it is unavailable through my credentials[1]. Nevertheless, phylogenetic branching of Zika virus shows how the distinct Asian lineage spread into America[2]. Because AF372422.1 is the top BLAST hit and clusters more closely with Latin American strains, including Brazilian ones, than with Asian strains in Fig. 1, it likely represents the patient's isolate. This aligns with the patient's probable infection source in Brazil, given his recent trip.

On the other hand, Fig. 2 presents the phylogenetic tree of the broader query. It has a wider geographic range, now including Africa in addition to Latin America and Asia. This aligns with the known African and Asian lineages of Zika virus, where the Asian lineage (emerging from Africa) drove the global spread and major outbreaks. Early in the 20th century, Zika arose in Uganda before spreading to West Africa and Asia[1, 3]. Note that AF372422.1 is absent from Fig. 2 tree. This might be due to GenBank's specificity in search results (see the first paragraph).

Original Antigenic Sin role in Zika virus is particularly relevant due to its close evolutionary relationship with other viruses, such as Dengue. Zika and Dengue virus are evolutionarily related and share a common ancestry within the Aedes-borne flaviviruses[4]. This leads to cross-reactive immune responses. When a patient contracts Zika, the immune system tends to produce antibodies that are more cross-reactive in individuals with prior dengue exposure than in those who have never had dengue. These cross-reactive antibodies have low neutralizing potential against Zika virus and can potentially enhance Zika infection through antibody-dependent enhancement[5].

Conclusion

The Fig. 1 phylogenetic tree shows a predominance of Latin American strains, suggesting the patient's strain is related to Brazilian's, consistent with its likely infection source. Asian strains also appear, reflecting Zika's evolutionary history. Fig. 2 supports this by highlighting the virus's relationship to African lineages, aligning with known Zika phylogeny. Finally, understanding how Zika interacts with related flaviviruses such as Dengue is crucial for public health, as prior flavivirus immunity may influence susceptibility to severe disease.

Methodology

I completed steps 1-4 of the assignment. I used iTOL in step 5 to visualize my phylogenetic tree online[6]. To color strains by country, I created a metadata file mapping each strain to its origin and a specific color. I programmed three R scripts: one to extract strain IDs from the MUSCLE alignment output, another to retrieve metadata via rentrez, and a third to generate the iTOL-compatible TXT file. Documentation and higher resolution figures are available[7].

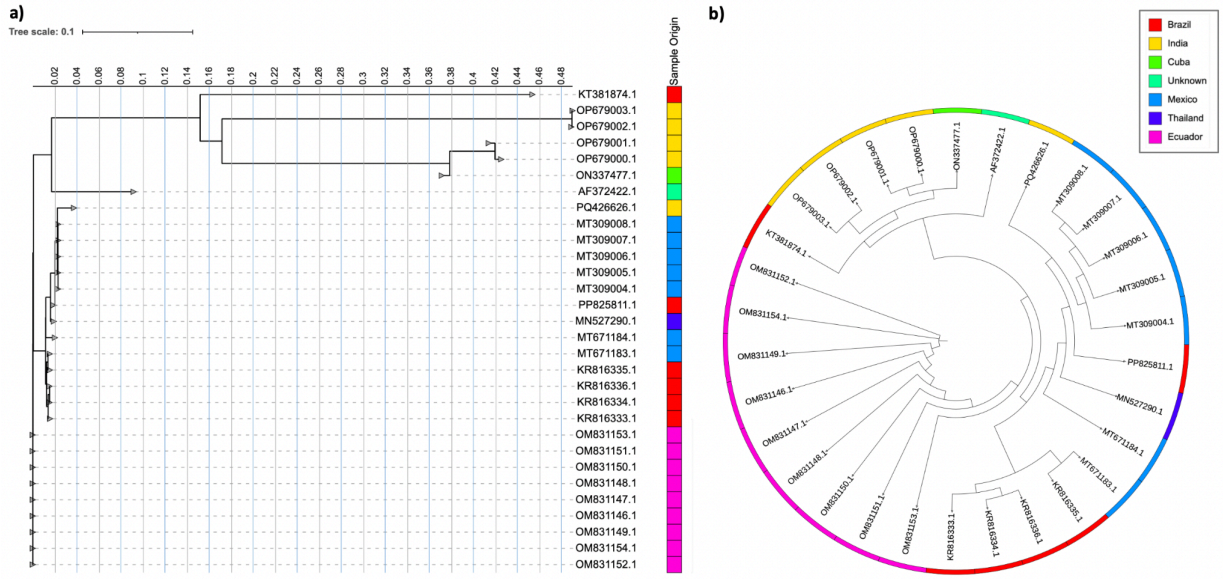


Figure 1. Phylogenetic Trees constructed from 30 sequences retrieved from the Genbank query "Zika virus envelope protein (E) gene". Strains are color-coded by country of origin. a) Rectangular phylogenetic tree with branch lengths proportional to evolutionary distance. b) Circular phylogenetic tree, where branch lengths are not considered.

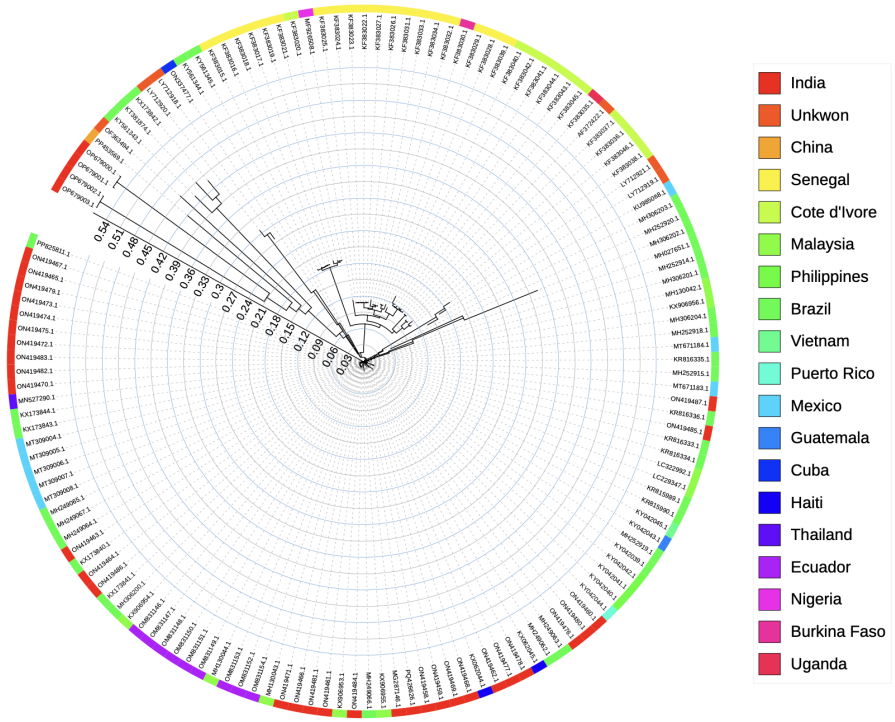


Figure 2. Circular Phylogenetic tree constructed from 146 retrieved from Genbank query "Zika virus envelope protein". Strains are color-coded by country of origin. Tree branch lengths are proportional to evolutionary distance.

References

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